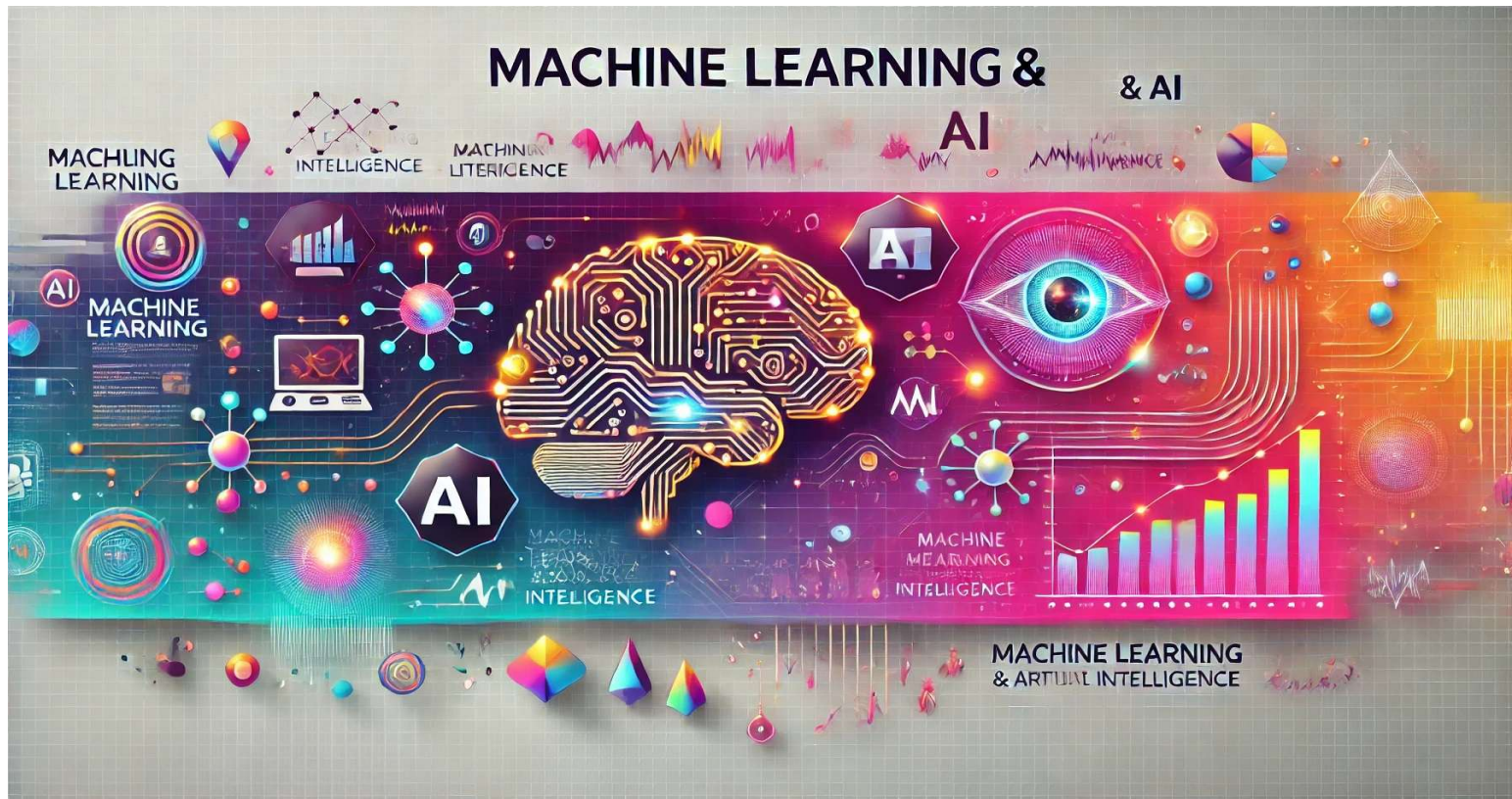




Python with Machine Learning 6-Month Full Syllabus

Module 1: Python Programming (Month 1)

✓ Python Basics

- Installation (Anaconda, Jupyter Notebook, VS Code)
 - Variables & Data Types
 - (Strings, Lists, Tuples, Sets, Dictionaries)
 - Operators (Arithmetic, Logical, Bitwise, Membership)
 - Control Flow (If-Else, Loops – for, while, nested)
 - Functions (Built-in, User-Defined, Lambda Functions)
-

✓ Object-Oriented Programming (OOPs)

- Classes & Objects
 - Encapsulation, Inheritance, Polymorphism
 - Magic Methods & Dunder Methods
 - File Handling (Reading/Writing Text, CSV, JSON Files)
-

✓ Exception Handling & Logging

- Try-Except Blocks
 - Raising Exceptions
 - Logging Errors
-

✓ Python Libraries Overviews

- NumPy,
 - Pandas,
 - Matplotlib,
 - Seaborn,
 - Scikit-learn
-

Module 2: Data Handling & Processing (Month 2)

✓ NumPy for Numerical Computing

- Arrays, Array Indexing & Slicing
 - Mathematical & Statistical Operations
 - Broadcasting & Vectorization
 - Linear Algebra (Dot Product, Matrix Multiplication)
-

✓ Pandas for Data Analysis

- DataFrames & Series
 - Handling Missing Data
 - Data Cleaning & Transformation
 - Data Aggregation (GroupBy, Pivot Tables)
-

✓ Data Visualization with Matplotlib & Seaborn

- Line, Bar, Histogram, Scatter, Pie Charts
 - Boxplots, Violin Plots, Pairplots
 - Heatmaps for Correlation Analysis
-

✓ Exploratory Data Analysis (EDA)

- Identifying & Handling Outliers (Z-Score, IQR)
 - Feature Engineering (Creating New Features, Transformations)
 - Handling Categorical Data (Label Encoding, One-Hot Encoding)
-

✓ SQL for Data Science

- Basic SQL Commands (DDL, DML, DQL)
 - Joins, Subqueries, Window Functions
 - Writing SQL Queries for Data Extraction
-

Module 3: Machine Learning Foundations (Month 3)

This module introduces core machine learning concepts and algorithms.

Topics Covered:

✓ Introduction to Machine Learning

- Types of ML: Supervised, Unsupervised, Reinforcement Learning
 - Understanding Model Evaluation Metrics (Accuracy, Precision, Recall, F1-Score)
 - Bias-Variance Tradeoff
-

✓ Regression Models (Predicting Continuous Values)

- Simple & Multiple Linear Regression
 - Assumptions of Linear Regression
 - Model Interpretation (R-Squared, Adjusted R-Squared)
-

✓ Classification Models (Predicting Categories)

- Logistic Regression (Binary & Multiclass)
 - ROC Curve, Precision-Recall Curve, Confusion Matrix
-

✓ Feature Selection & Dimensionality Reduction

- Feature Selection Techniques (Variance Threshold, SelectKBest, RFE)
 - Principal Component Analysis (PCA)
 - Singular Value Decomposition (SVD)
-

Module 4: Advanced Machine Learning (Month 4)

This module dives into more advanced machine learning techniques.

Topics Covered:

✓ Decision Trees & Random Forest

- Gini Impurity, Entropy
 - Over fitting & Pruning
 - Random Forest Algorithm
-

✓ Support Vector Machines (SVM)

- Kernel Trick (Linear, Polynomial, RBF)
 - Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV)
-

✓ Ensemble Learning Techniques

- Bagging (Bootstrap Aggregation)
 - Boosting (AdaBoost, Gradient Boosting, XGBoost, CatBoost)
 - Stacking & Blending
-

✓ Handling Imbalanced Datasets

- SMOTE (Synthetic Minority Over-sampling)
 - Class Weights
 - Threshold Adjustments
-

✓ hyper parameter Tuning & Optimization

- GridSearchCV & Randomized SearchCV
 - Cross-validation
 - ---
-

Module 5: Unsupervised Learning & Advanced Techniques (Month 5)

✓ Clustering Algorithms

- K-Means Clustering (Choosing K using Elbow Method)
 - Hierarchical Clustering (Agglomerative, Divisive)
 - DBSCAN Algorithm
-

✓ Natural Language Processing (NLP)

- Text Preprocessing (Tokenization, Lemmatization, Stopwords Removal)
 - TF-IDF & Word Embeddings (Word2Vec, GloVe)
 - Sentiment Analysis & Text Classification
-

✓ Time Series Analysis

- Time Series Components (Trend, Seasonality, Noise)
 - Moving Averages, ARIMA, SARIMA Models
 - Forecasting with Machine Learning
 - Reinforcement Learning Basics
 - Markov Decision Process (MDP)
 - Q-Learning, Deep Q-Networks (DQN)
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Module 6: Deep Learning & Deployment (Month 6)

✓ Neural Networks (ANN & CNN)

- Perceptron, Multi-layer Perceptron (MLP)
 - Activation Functions (ReLU, Sigmoid, Softmax)
 - Forward & Backpropagation
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✓ Convolution Neural Networks (CNN)

- Convolutions, Pooling, Feature Extraction
 - Transfer Learning (Using Pretrained Models)
-

✓ Recurrent Neural Networks (RNN) & LSTMs

- Handling Sequential Data
 - Stock Price Prediction Using LSTMs
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✓ Model Deployment

- Deploying ML Models using Flask & FastAPI
 - Creating Web Apps using Streamlit
-

Hands-on Projects throughout the Course

Regression: House Price Prediction

Classification: Customer Churn Prediction

Clustering: Customer Segmentation for E-commerce

NLP: Sentiment Analysis on Amazon Reviews

Time Series: Forecasting Stock Prices

Deep Learning: Image Classification using CNN

Learning Outcome:

By the end of this course, you will be able to:

✓ Build end-to-end ML pipelines from data collection to model deployment.

✓ Work with real-world datasets and implement ML models for various business cases.

- ✓ Optimize and tune models using cross-validation & hyperparameter tuning.
- ✓ Deploy machine learning models using Flask, FastAPI & Streamlit.
- ✓ Gain expertise in deep learning frameworks (TensorFlow & Keras).

This in-depth module-based syllabus ensures you are industry-ready with hands-on experience in Machine Learning and Deep Learning.
